

5	(a)	Current = $2.7 - 1.5 = 1.2 \text{ A}$	B1
	(b)	p.d. across XY = $12 \text{ V} = 1.5 (5.0 + R)$ Resistance R = 3.0Ω	B1
	(c)	p.d. across XZ = $(1.6/2.0)12 = 9.6 \text{ V}$ p.d. across XW = $1.5(5.0) = 7.5 \text{ V}$ potential difference = $9.6 - 7.5 = 2.1$	C1 A1
	(d)	When the resistance of the variable resistor is now increased, the <u>effective resistance of the circuit increases</u> . For the same e.m.f. supply, the <u>power dissipated in the effective resistor decreases</u> since <u>power is inversely proportional to the effective resistance</u> . Thus the power supplied by the battery <u>decreases</u> . OR When the resistance of the variable resistor is now increased, the <u>effective resistance of the 5.0 ohm and variable resistor increases</u> , so <u>the current in the variable resistor decreases</u> . The <u>current in the resistance wire is unchanged</u> since the p.d. across the wire and the resistance remain the same. So, <u>the current in the battery decreases (same e.m.f.)</u> <u>the power decreases</u> .	B1 B1 B1 (B1) (B1) (B1)

6	(a)		
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